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CSC 101 — Past Questions & Materials

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Course: CSC 101

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MODULE 1

Basic Computing Concepts

A **computer** can be described as an electronic device that accepts data as input, processes the data based on a set of predefined instructions called program to produce the result of these operations as output called information. From this description, a computer can be referred to as an Input-Process-Output (IPO) system, pictorially represented in the Figure 1:

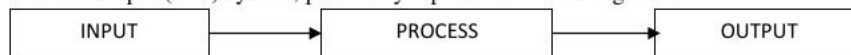


Figure 1.1: IPO Representation of a computer System

Data are raw facts, such as a score in examination or the name of a student, for example 55 or Malik respectively. There are three types of data – Numeric, alphabetic, and alphanumeric. Numeric data consists of digits 0 – 9 (such as 31), while alphabetic data consist of any of the English language alphabets in upper and lower cases (e. g. Toyin). An alphanumeric data can consist of a number, an alphabet or a special character, such as a vehicle plate number (e. g. AE 731 LRN).

Information: data as described above contain no meaning, however, when it is transformed into a more meaningful and useful form, it is called **information**. The transformation process involves a series of operations to be performed by the computer on the raw data that are fed into the system. The operation can be arithmetic (such as addition, subtraction, multiplication, and division), logical comparison or character manipulation (as in text processing).

Logical comparison means testing whether one data item is greater than, equal to, or less than another item, and based on the outcome of the comparison, a specified action can be taken. The output of the processing can be in form of reports which can be displayed or printed.

The History of Computer

In the early days of man, fingers and toes were used for counting. Later on, sticks and pebbles were used. Permanent records of the result of counting were kept by putting marks on the ground, wall and so on using charcoal, chalk, and plant juice.

The historical development of computing focuses on the digital computer from the **Abacus** to the modern **electronic computer**. Some of these people whose contributions have been widely acknowledged to the development of Computer will be discussed:

Abacus

The abacus was invented to replace the old methods of counting. It is an instrument known to have been used for counting as far back as 500 B.C. in Europe, China, Japan and India and it is still being used in some parts of China today.

The abacus qualifies as a digital instrument because it uses beads as counter to calculate in discrete form. It is made of a board that consists of beads that slide on wires. The abacus is divided by a wooden bar or rod into two zones. Perpendiculars to this rod are wires arranged in parallel, each one representing a positional value. Each zone is divided into two levels - upper and lower. Two

beads are arranged on each wire in the upper zone, while five beads are arranged on each wire in the lower zone.

The abacus can be used to perform arithmetic operations such as addition and subtraction efficiently.



Figure 1.2: Modern abacus.

Note that the abacus is really just a representation of the human fingers: the 5 lower rings on each rod represent the 5 fingers and the 2 upper rings represent the 2 hands.

Blaise Pascal

Pascal was born at Clermont, France in 1623 and died in Paris in 1662. Pascal was a Scientist as well as a Philosopher. He started to build his mechanical machine in 1640 to aid his father in calculating taxes. He completed the first model of his machine in 1642 and it was presented to the public in 1645.

The machine, called Pascal machine or Pascaline, was a small box with eight dials that resembled the analog telephone dials. Each dial is linked to rotating wheel that displayed the digits in a register window. Pascal's main innovative idea was the linkage provided for the wheels such that an arrangement was made for a carry from one wheel to its left neighbour when the wheel passed from a display of 9 to 0. The machine could add and subtract directly.

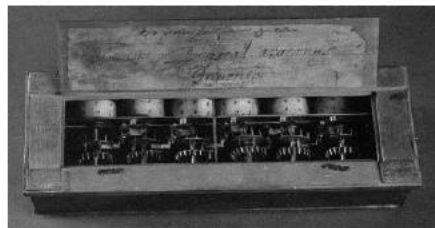


Figure 1.3: Pascal's Pascaline [photo © 2002 IEEE]

A Pascaline opened up so you can observe the gears and cylinders which rotated to display the numerical result

Joseph Marie Jacquard

In 1801 the Frenchman Joseph Marie Jacquard invented a power loom that could base its weave (and hence the design on the fabric) upon a pattern automatically read from punched wooden cards,

held together in a long row by rope. Descendents of these *punched cards* have been in use ever since.



Figure 1.4: Jacquard's Loom showing the threads and the punched cards

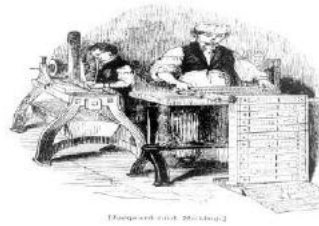


Figure 1.5:By selecting particular cards for Jacquard's loom you defined the woven pattern
[photo © 2002 IEEE]

Charles Babbage

Charles Babbage was born in Totnes, Devonshire on December 26, 1792 and died in London on October 18, 1871. He was educated at Cambridge University where he studied Mathematics. In 1828, he was appointed Lucasian Professor at Cambridge. Charles Babbage started work on his analytic engine when he was a student. His objective was to build a program-controlled, mechanical, digital computer incorporating a complete arithmetic unit, store, punched card input and a printing mechanism.

The program was to be provided by the set of Jacquard cards. However, Babbage was unable to complete the implementation of his machine because the technology available at his time was not adequate to see him through. Moreover, he did not plan to use electricity in his design. It is noteworthy that Babbage's design features are very close to the design of the modern computer. Babbage invented the modern postal system, cowcatchers on trains, and the ophthalmoscope, which is still used today to treat the eye.

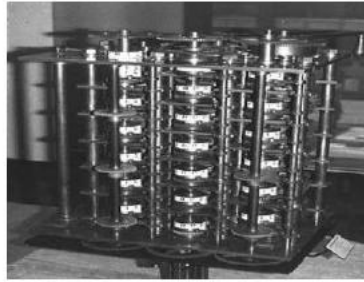


Figure1.6: A small section of the type of mechanism employed in Babbage's Difference Engine [photo © 2002 IEEE]

Augusta Ada Byron

Ada Byron was the daughter of the famous poet Lord Byron and a friend of Charles Babbage, (Ada later become the Countess Lady Lovelace by marriage). Though she was only 19, she was fascinated by Babbage's ideas and through letters and meetings with Babbage she learned enough about the design of the Analytic Engine to begin fashioning programs for the still unbuilt machine. While Babbage refused to publish his knowledge for another 30 years, Ada wrote a series of "Notes" wherein she detailed sequences of instructions she had prepared for the Analytic Engine. The Analytic Engine remained unbuilt but Ada earned her spot in history as the first computer programmer. Ada invented the subroutine and was the first to recognize the importance of looping.

Herman Hollerith

Hollerith was born at Buffalo, New York in 1860 and died at Washington in 1929. Hollerith founded a company which merged with two other companies to form the Computing Tabulating Recording Company which in 1924 changed its name to International Business Machine (IBM) Corporation, a leading company in the manufacturing and sales of computer today.

Hollerith, while working at the Census Department in the United States of America became convinced that a machine based on cards can assist in the purely mechanical work of tabulating population and similar statistics was feasible. He left the Census in 1882 to start work on the Punch Card Machine which is also called Hollerith desks.

This machine system consisted of a punch, a tabulator with a large number of clock-like counters and a simple electrically activated sorting box for classifying data in accordance with values punched on the card. The principle he used was simply to represent logical and numerical data in the form of holes on cards.

His system was installed in 1889 in the United States Army to handle Army Medical statistics. He was asked to install his machine to process the 1890 Census in USA. This he did and in two years, the processing of the census data was completed which used to take ten years. Hollerith's machine was used in other countries such as Austria, Canada, Italy, Norway and Russia.



Figure 1.7: Hollerith desks [photo courtesy The Computer Museum]

John Von Neumann

Von Neumann was born on December 28, 1903 in Budapest, Hungary and died in Washington D. C. on February 8, 1957. He was a great mathematician with significant contribution to the theory of games and strategy, set theory and the design of high speed computing machines. In 1933, he was appointed one of the first six professors of the school of mathematics in the institute for Advanced Study at the Princeton University, USA, a position he retained until his death.

Neumann with some other people presented a paper titled “The Preliminary discussion of the Logical Design of an Electronic Computing Instrument” popularly known as Von Neumann machine. This paper contains revolutionary ideas on which the present-day computers are based. The machine has Storage, Control, Arithmetic and input/output units. The machine was to be a general-purpose computing machine. It was to be an electronic machine and introduced the concept of stored program. This concept implied that the operations in the computer were to be controlled by a program stored in the memory of the computer. This program was to consist of codes that intermixed data with instructions.

As a result of this, it became possible for computations to proceed at electronic speed, perform the same set of operations or instructions repeatedly and the concept of program counter, which implied that whenever an instruction is fetched, the program counter which is a high-speed register automatically contains the address of the instruction to be executed next.

J. V. Atanasoff

One of the earliest attempts to build an all-electronic digital computer occurred in 1937 by J. V. Atanasoff, a professor of physics and mathematics at Iowa State University. By 1941 he and his graduate student, Clifford Berry, had succeeded in building a machine that could solve 29 simultaneous equations with 29 unknowns. This machine was the first to store data as a charge on a capacitor, which is how today computers stored information is in their main memory. It was also the first to employ binary arithmetic. However, the machine was not programmable, it lacked a conditional branch, its design was appropriate for only one type of mathematical problem, and it was not further pursued after World War II.



Figure 1.8: The Atanasoff-Berry Computer [photo © 2002 IEEE]

Howard Aiken

Howard Aiken of Harvard was the principal designer of the Mark I. The Harvard Mark I computer was built as a partnership between Harvard and IBM in 1944. This was the first programmable digital computer made in the U.S. But it was not a purely electronic computer. Instead the Mark I was constructed out of switches, relays, rotating shafts, and clutches. The machine weighed 5 tons, incorporated 500 miles of wire, was 8 feet tall and 51 feet long, and had a 50ft rotating shaft running its length, turned by a 5 horsepower electric motor. The Mark I ran non-stop for 15 years.



Figure 1.9: The Harvard Mark I: An electro-mechanical computer

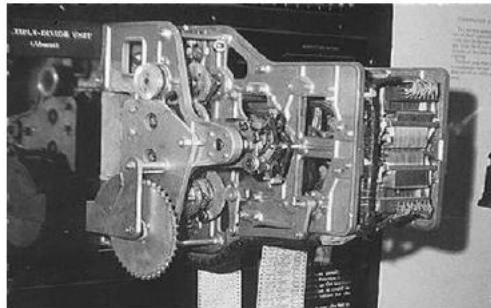


Figure 1.10: One of the four paper tape readers on the Harvard Mark I

Grace Hopper

Grace Hopper was one of the primary programmers for the Mark I. Hopper found the first computer "bug": a dead moth that had gotten into the Mark I and whose wings were blocking the reading of the holes in the paper tape. The word "bug" had been used to describe a defect since at least 1889 but Hopper is credited with coining the word "debugging" to describe the work to eliminate program faults.

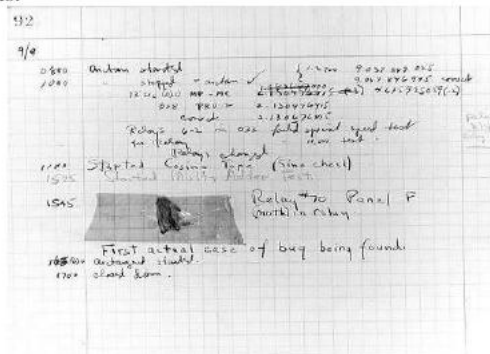


Figure 1.11: The first computer bug [photo © 2002 IEEE]

In 1953 Grace Hopper invented the first high-level language, "Flow-matic". This language eventually became COBOL which was the language most affected by the infamous Y2K problem. A high-level language is designed to be more understandable by humans than is the binary language understood by the computing machinery. A high-level language is worthless without a program -- known as a compiler -- to translate it into the binary language of the computer and hence Grace Hopper also constructed the world's first compiler. Grace remained active as a Rear Admiral in the Navy Reserves until she was 79.

Bill Gates

William (Bill) H. Gates was born on October 28, 1955 in Seattle, Washington, USA. Bill Gates decided to drop out of college so he could concentrate all his time writing programs for Intel 8080 categories of Personal Computers (PC). This early experience put Bill Gates in the right place at the right time once IBM decided to standardize on the Intel microprocessors for their line of PCs in 1981. Gates founded a company called Microsoft Corporation (together with Paul G. Allen) and released its first operating system called MS-DOS 1.0 in August, 1981 and the last of its group in (MS-DOS 6.22) April, 1994. Bill Gates announced Microsoft Windows on November 10, 1983.

Philip Emeagwali

Philip Emeagwali was born in 1954, in the Eastern part of Nigeria. He had to leave school because his parents couldn't pay the fees and he lived in a refugee camp during the civil war. He won a scholarship to university. He later migrated to the United States of America. In 1989, he invented the formula that used 65,000 separate computer processors to perform 3.1 billion calculations per second.

Philip Emeagwali, a supercomputer and Internet pioneer is regarded as one of the fathers of the internet because he invented an international network which is similar to, but predates that of the

Internet. He also discovered mathematical equations that enable the petroleum industry to recover more oil. Emeagwali won the 1989 Gordon Bell Prize, computation's Nobel Prize, for inventing a formula that lets computers perform the fastest computations, a work that led to the reinvention of supercomputers.

MODULE 2

Basic Component of Computer

Components of Computer refer to physical and the non-physical part of the system. A computer system can be divided into **hardware, software and humanware**

The Hardware

The hardware refers to the physical components and the devices which make up the visible computer. It can be divided into two: **Central Processing Unit (CPU)** and the **Peripherals**. The **CPU** is responsible for all processing that the computer does while the peripherals are responsible for feeding data into the system and for collecting information from the system.

The CPU consists of **Main storage, ALU and Control Unit**. The **main storage** is used for storing data to be processed as well as the instructions for processing them. The **ALU** is the unit for arithmetic and logical operations. The **control unit** ensures the smooth operation of the other hardware units. It fetches instruction, decode (interprets) the instruction and issues commands to the units responsible for executing the instructions.

The **peripherals** are in three categories: **Input devices, Output devices and auxiliary storage devices**.

The **input** device is used for supplying data and instructions to the computer. Examples are terminal Keyboard, Mouse, Joystick, Microphone, Scanner, Webcam, etc.

Output device is used for obtaining result (information) from the computer. Examples are Printers, Video Display Unit (VDU), loudspeaker, projector etc,

Auxiliary Storage Devices are used for storing information on a long-term basis. Examples are hard disk, flash disk, magnetic tape, memory card, solid state drive SDD etc.

A simple model of the hardware part of a computer system is shown below:

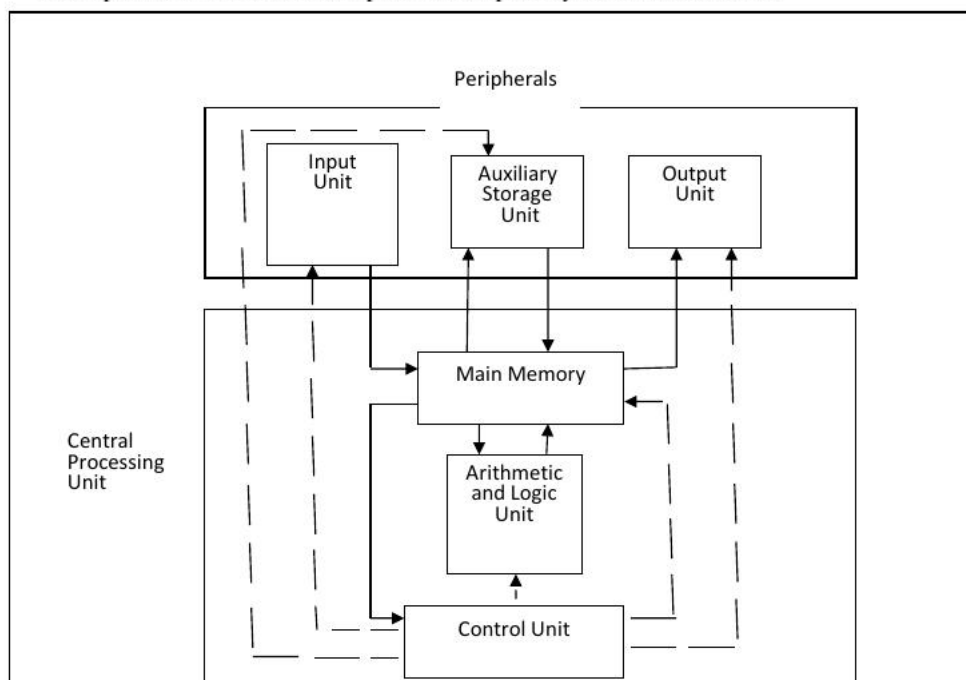


Figure 2.1: Hardware part of a computer system

1.1 The Software

Software are basically referred to as programs. A **program** consists of sequence of instructions required to accomplish well-defined tasks. Examples of such tasks include:

1. Finding the average score of a student
2. Computing the net pay of an employee
3. Solving a set of simultaneous linear equations

It is the software that enables the hardware to be put into effective use. There are two main categories of software – **System software** and **Application software**.

3.3.1 System Software

System software are programs commonly written by computer manufacturers, which have direct effect on the control, performance and ease of usage of the computer system. Examples are Operating System, Language Translators, Utilities and Service Programs, and Database Management Systems (DBMS).

Operating System is a collection of program modules which form an interface between the computer hardware and the computer user. Its main function is to ensure a judicious and efficient utilization of all the system resources (such as the processor, memory, peripherals and other system data) as well as to provide programming convenience for the user. Examples are Unix, Linux, Windows, Macintosh, and Disk Operating system.

Language Translators are programs which translate programs written in non-machine languages such as FORTRAN, C, Pascal, and BASIC into the machine language equivalent. Example of language translators are assemblers, interpreters, compilers and preprocessor.

- **Assemblers:** This is a program that converts program written in assembly language (low level language) into machine language equivalent.
- **Interpreter:** This is a program that converts program written in high level language (HLL) into its machine language (ML) equivalent one line at a time. Language like BASIC is normally interpreted.
- **Compiler:** This is a program that translates program written in high level language (HLL) into machine language (ML) equivalent all at once. Compilers are normally called by the

names of the high-level language they translate. For instance, COBOL compiler, FORTRAN compiler etc.

- **Preprocessor:** This is a language translator that takes a program in one HLL and produces equivalent program in another HLL. For example, there are many preprocessors to map structured version of FORTRAN into conventional FORTRAN.

Database Management System (DBMS) is a complex program that is used for creation, storage, retrieving, securing and maintenance of a database. A **database** can be described as an organized collection of related data relevant to the operations of a particular organization. The data are stored usually in a central location and can be accessed by different authorized users.

Linker is a program that takes several object files and libraries as input and produces one executable object file.

Loader is a program that places an executable object file into memory and makes them ready for execution. Both linker and loader are provided by the operating system.

Utility and Service Programs

These are programs which provide facilities for performing common computing tasks of a routine nature. The following are some of the examples of commonly used utility programs:

- **Sort Utility:** This is used for arranging records of a file in a specified sequence (alphabetic, numerical or chronological) of a particular data item within the records. The data item is referred to as the sort key.
- **Merge Utility:** This is used to combine two or more already ordered files together to produce a single file.
- **Copy Utility:** This is used mainly for transferring data from a storage medium to the other, for example from disk to tape.
- **Debugging Facilities:** These are used for detecting and correcting errors in program.
- **Text Editors:** These provide facilities for creation and amendment of program from the terminal.
- **Benchmark Program:** This is a standardized collection of programs that are used to evaluate hardware and software. For example, a benchmark might be used to compare the performance of two different computers on identical tasks, assess the comparative performance of two operating systems etc.

Application Software

These are programs written by a user to solve individual application problem. They do not have any effect on the efficiency of the computer system. An example is a program to calculate the grade point average of all the 100L students. Application software can be divided into two namely: **Application Package** and **User's Application Program**. When application programs are written in a very generalized and standardized nature such that they can be adopted by a number of

different organizations or persons to solve similar problem, they are called **Application Packages**. There are a number of micro-computer based packages. These include word processors (such as Ms-word, WordPerfect, WordStar); Database packages (such as Oracle, Ms-access, Sybase, SQL Server, and Informix); Spreadsheet packages (such as Lotus 1-2-3 and Ms-Excel); Graphic packages (such as CorelDraw, Fireworks, Photoshop etc), and Statistical packages (such as SPSS). **User's Application Program** is a program written by the user to solve specific problem which is not generalized in nature. Examples include writing a program to find the roots of quadratic equation, payroll application program, and program to compute students' results.

The Human-ware

Although, the computer system is automatic in the sense that once initiated, it can, without human intervention, continue to work on its own under the control of stored sequence of instructions (program), however, it is not automatic in the sense that it has to be initiated by a human being, and the instructions specifying the operations to be carried out on the input data are given by human being. Therefore, apart from the hardware and software, the third element that can be identified in a computer system is the **human-ware**. This term refers to the people that work with the computer system. The components of the human-ware in a computer system include the system analyst, the programmer, data entry operator, end users etc.

Organizational Structure of a Typical Computer Installation

The following diagram shows the organizational structure of a typical computer installation

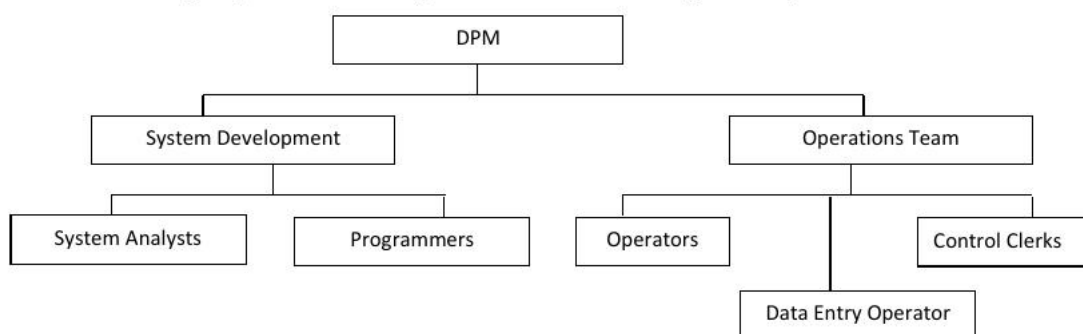


Figure 2.2: Organizational Structure of a typical computer installation

Data Processing Manager (DPM) supervises every other persons that work with him and is answerable directly to the management of the organization in which he works.

A System Analyst is a person that understudies an existing system in operation in order to ascertain whether or not computerization of the system is necessary and/or cost-effective. When found necessary, he designs a computerized procedure and specifies the functions of the various programs needed to implement the system.

A Programmer is the person that writes the sequence of instructions to be carried out by the computer in order to accomplish a well-defined task. The instructions are given in computer programming languages.

A data entry operator is the person that enters data into the system via keyboard or any input device attached to a terminal. There are other ancillary staffs that perform other functions such as controlling access to the computer room, controlling the flow of jobs in and out of the computer room.

An end-user is one for whom a computerized system is being implemented. The end-user interacts with the computerized system in their day-to-day operations of the organization. For example a cashier in the bank who receives cash from customers or pays money to customers interacts with the banking information system.

MODULE 3

Boolean Algebra, Fundamentals of Truth tables and Precedence

Algebra

Algebra means reunion on broken parts. It is the study of mathematical symbols and rules for manipulating the symbols. Algebra can be regarded as elementary, abstract or modern depending on the level or field of study.

Algebra has computations similar to arithmetic but with letters standing for numbers which allows proofs of properties that are true regardless of the numbers involved. For example, quadratic equation: $ax^2 + bx + c = 0$ where a, b, c can be any number ($a \neq 0$). Algebra is used in many studies, for example, elementary algebra, linear algebra, Boolean algebra, and so on.

1.1 Polynomials

A polynomial involves operations of addition, subtraction, multiplication, and non-negative integer exponents of terms consisting of variables and coefficients. For example, $x^2 + 2x - 3$ is a polynomial in the single variable x . Polynomial can be rewritten using commutative, associative and distributive laws.

An important part of algebra is the factorization of polynomials by expressing a given polynomial as a product of other polynomials that cannot be factored any further. Another important part of algebra is computation of polynomial greatest common divisors. $x^2 + 2x - 3$ can be factored as $(x - 1)(x + 3)$.

1.2 Boolean Algebra

Boolean algebra is the branch of algebra in which the values of the variables are true values denoted by 1 and 0 or true and false respectively.

Boolean algebra can be used to describe logic circuit; it is also use to reduce complexity of digital circuits by simplifying the logic circuits. Boolean algebra is also referred to as Boolean logic. It was developed by George Boole sometime on the 1840s and is greatly used in computations and in computer operations. The name Boolean comes from the name of the author.

Boolean algebra is a logical calculus of truth values. It somewhat resembles the arithmetic algebra of real numbers but with a difference in its operators and operations. Boolean operations involve the set $\{0, 1\}$, that is, the numbers 0 and 1. Zero [0] represents “false” or “off” and One [1] represents “true” or “on”.

1 – True, on

0 – False, off

This has proved useful in programming computer devices, in the selection of actions based on conditions set.

Basic Boolean operations

1. AND

The AND operator is represented by a period or dot in-between the two operands e.g
- $X.Y$

The Boolean multiplication operator is known as the *AND* function in the logic domain; the function evaluates to 1 only if both the independent variables have the value 1.

2. OR

The OR operator is represented by an addition sign. Here the operation $+$ is different from that defined in normal arithmetic algebra of numbers. E.g. $X+Y$

The $+$ operator is known as the *OR* function in the logic domain; the function has a value of 1 if either or both of the independent variables has the value of 1.

3. NOT

The NOT operator is represented by X' or $\bar{X}$.

This operator negates whatever value is contained in or assigned to X . It changes its value to the opposite value. For instance, if the value contained in X is 1, X' gives 0 as the result and if the value stored in X is 0, X' gives 1 as the result. In some texts, NOT may be represented as $\bar{X}$

To better understand these operations, **truth table** is presented for the result of any of the operations on any two variables.

Truth Tables

A truth table is a mathematical table used in logic to compute the functional values of logical expressions on each of their functional arguments. It is specifically in connection with Boolean algebra and Boolean functions. Truth tables can be used to tell if a proposition expression is logically valid. In a truth table, the output is completely dependent on the input. It is composed of a column for each input entry and another column the corresponding output. Each row of the truth table therefore contains one possible configuration of the input variables (for instance, $X=\text{true}$ $Y=\text{false}$), and the result of the operation for those values.

Applications of truth table

1. The truth table can be used in analyzing arguments.
2. It is used to reduce basic Boolean operations in computing
3. It is used to test the validity of statements. In validating statements, the following three steps can be followed:
 - a. Represent each premise (represented as inputs) with a symbol (a variable).

- b. Represent the conclusion (represented as the final result) with a symbol (a variable).
- c. Draw a truth table with columns for each premise (input) and a column for the conclusion (result).

Truth tables are a means of representing the results of a logic function using a table. They are constructed by defining all possible combinations of the inputs to a function in the Boolean algebra, and then calculating the output for each combination in turn. The basic truth table shows the various operators and the result of their operations involving two variables only. More complex truth tables can be built from the knowledge of the foundational truth table. The number of input combinations in a Boolean function is determined by the number of variables in the function and this is computed using the formula 2^n .

Number of input combinations = 2^n . Where n is number of variable(s).

For example, a function with two variables has an input combination of $2^2=4$. Another with three variables has $2^3=8$ input combinations, and so on.

AND

X	Y	X . Y
0	0	0
0	1	0
1	0	0
1	1	0

OR

X	Y	X+Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT

X	X'
0	1
1	0

The NOT operation is a unary operator; it accepts only one input.

Example:

• Draw a truth table for $A+BC$.				
A	B	C	BC	A+BC
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

• Draw a truth table for $AB+BC$.					
A	B	C	AB	BC	AB+BC
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	1	1
1	0	0	0	0	1
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	1	1	1

• Draw a truth table for $A(B+D)$.				
A	B	D	B+D	A(B+D)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

$$J = f(A,B,C) = A\bar{B}\bar{C} + \bar{A}B\bar{C}$$

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$A\bar{B}\bar{C}$	$\bar{A}B\bar{C}$	$A\bar{B}\bar{C} + \bar{A}B\bar{C}$
0	0	0	1	1	1	0	1	1
0	0	1	1	1	0	0	0	0
0	1	0	1	0	1	0	0	0
0	1	1	1	0	0	0	0	0
1	0	0	0	1	1	1	0	1
1	0	1	0	1	0	0	0	0
1	1	0	0	0	1	0	0	0
1	1	1	0	0	0	0	0	0

Basic Logic Gates

Logic can be viewed as black boxes with binary input (independent variable) and binary output (dependent variable). It also refers to both the study of modes of reasoning and the use of valid reasoning. In the latter sense, logic is used in most intellectual activities. Logic in computer science has emerged as a discipline and it has been extensively applied in the fields of Artificial Intelligence, and Computer Science, and these fields provide a rich source of problems in formal and informal logic.

Boolean logic, which has been considered as a fundamental part to computer hardware, particularly, the system's arithmetic and logic structures, relating to operators AND, NOT, and OR.

Logic gates

A logic gate is an elementary building block of a digital circuit. Complex electronic circuits are built using the basic logic gates. At any given moment, every terminal of the logic gate is in one of the two binary conditions low (0) or high (1), represented by different voltage levels.

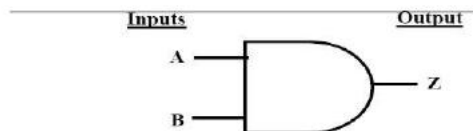
There are 3 basic logic gates: AND, OR, NOT.

Other gates- NAND, NOR, XOR and XNOR are based on the 3 basic gates.

The AND gate

The AND gate is so called because, if 0 is called "false" and 1 is called "true," the gate acts in the same way as the logical "and" operator. The following illustration and table show the circuit symbol and logic combinations for an AND gate.

The output is "true" when both inputs are "true." Otherwise, the output is "false."



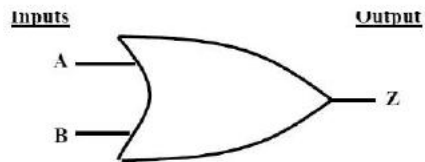
Logic Function: $Z = AB$

Truth Table:

Inputs		Output
A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

The OR gate

The OR gate gets its name from the fact that it behaves after the way of the logical "or." The output is "true" if either or both of the inputs are "true." If both inputs are "false," then the output is "false."



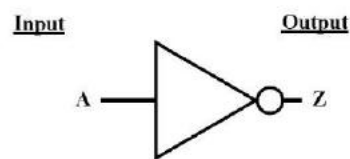
Logic Function: $Z = A + B$

Truth Table:

Inputs		Output
A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

The NOT gate

A logical inverter, sometimes called a NOT gate to differentiate it from other types of electronic inverter devices, has only one input. It reverses the logic state (i.e. its input).



Logic Function: $Z = \overline{A}$

Truth Table:

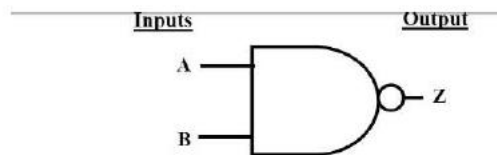
Input	Output
A	Z
0	1
1	0

As previously considered, the AND, OR and NOT gates' actions correspond with the AND, OR and NOT operators.

More complex functions can be constructed from the three basic gates by using DeMorgan's Law.

The NAND gate

The NAND gate operates as an AND gate followed by a NOT gate. It acts in the manner of the logical operation "and" followed by negation. The output is "false" if both inputs are "true." Otherwise, the output is "true". It finds the AND of two values and then finds the opposite of the resulting value.



Logic Function: $Z = \overline{AB}$

Truth Table:

Inputs		Output
A	B	Z
0	0	1
0	1	1
1	0	1
1	1	0

The NOR gate

The NOR gate is a combination of an OR gate followed by an inverter. Its output is "true" if both inputs are "false." Otherwise, the output is "false". It finds the OR of two values and then finds the complement of the resulting value.



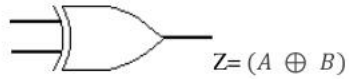
Logic Function: $Z = \overline{A + B}$

Truth Table:

Inputs		Output
A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

The XOR gate

The XOR (exclusive-OR) gate acts in the same way as the logical "either/or." The output is "true" if either, but not both, of the inputs are "true." The output is "false" if both inputs are "false" or if both inputs are "true." Another way of looking at this circuit is to observe that the output is 1 if the inputs are different, but 0 if the inputs are the same.

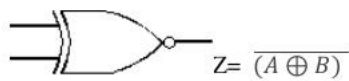


XOR gate

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

The XNOR gate

The XNOR (exclusive-NOR) gate is a combination of an XOR gate followed by an inverter. Its output is "true" if the inputs are the same, and "false" if the inputs are different. It performs the operation of an XOR gate and then inverts the resulting value.



XNOR gate

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	1

DIVERSE AND GROWING COMPUTERS AND DIGITAL APPLICATIONS.

Diverse And Growing Computers

"Diverse and Growing Computers" simply refers to the wide range of computing devices, systems and technologies that are constantly evolving to meet various needs and applications.

Similarly, It is the increasing representation of underrepresented groups (such as women, people of color, individuals with disabilities in the field of computer science and technology industry, and the development of more inclusive design teams and technologies.

Thus, Data diversity refers to the range of different types of elements in a dataset. The causes behind the growing importance of computer could be traced to their versatility, Efficiency, ability to process vast amounts of information at incredible speeds, crucial technology, highway and telecommunication services. Etc. computer save our valuable time in any work.

Diversity in computing is very important to drastically increase creativity, income, connection and untimely better projects that fit all sorts of people.

In this era, With the help of computer, Students better understand the basic concepts. Actively, including everyone and having diversity in computing brings an advantage due to the immense knowledge and perspectives of many different groups.

The diversity in tech can lead to better thinking, greater innovation, productivity and profit. It can also foster creativity as people with different backgrounds contribute to building technology.

Two issues that cause the lack of diversity are:

1. Pipeline: - The lack of early access to resources.
2. Culture: Exclusivity and discrimination in the workplace.

The lack of diversity can also be attributed to limited early exposure to resources, as students who do not already have computer skills upon entering college are at a disadvantage in computing majors. There is also the issue of discrimination and harassment faced in the workplace which affects all underrepresented groups. For example, studies have shown that 50% of women reported experiencing sexual harassment in tech companies.

Impacts of Enrollment Growth on Diversity in Computing, These showcase the diversity and growth in the computing field, spanning from hardware and software advancements to social impact and emerging technologies

Diversity is critical to success in any field. Diversity of perspectives and experiences results in robust thinking and approaches that can help yield solutions and products that meet the needs of a diverse customer base, which often improves the value of a product across the spectrum of

users. Diversity is often linked to positive outcomes, such as greater innovation, productivity, and profit.

A recent industry report identified a “massive economic opportunity” associated with increasing the ethnic and gender diversity in the Nigeria technology workforce, with the potential to add 470 to 570 billion to the Nigeria tech sector and support the creation of jobs and the improvement of products. The report identifies underrepresentation of African workers in the tech industry compared to the Nigeria workforce as a whole, accounting for 7 and 8 percent of tech workers, respectively, compared to 12 and 16 percent of all Nigeria workers.

The lack of diversity in computer science and in the information technology sector of the economy, especially among women and underrepresented minorities, is a well-recognized challenge. These representation rates are even smaller than those reported for the tech industry as a whole when considering diversity in computing occupations among all Nigeria institutions Bachelor’s degree holders.

The diverse and growing as it evolves in the following aspects of computing:

- (1) **Hardware:** Emerging Architectures (e.g., quantum computing, neuromorphic computing), Artificial Intelligence Processors (e.g., TPUs, GPUs), Internet of Things (IoT) Devices, Wearable Technology (e.g., smartwatches, fitness trackers), Autonomous Vehicles.
- (2) **Software:** These Includes: -Cloud Computing (e.g. Infrastructure as a Services, (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), Edge Computing, Artificial Intelligence (AI) and Machine Learning (ML), Blockchain and Distributed Ledger Technology, Virtual and Augmented Reality (VR/AR).
- (3) **Networking:** -5G and 6G Networks, Wi-Fi 6 and Future Wireless Standards, Network Function Virtualization (NFV), Software-Defined Networking (SDN), Internet of Bodies (IoB)

The Applications/Uses can be seen in different areas, such as Healthcare Technology (e.g., Telemedicine, Medical imaging), Financial Technology (FinTech), Cybersecurity and Threat Intelligence, Environmental Monitoring and Sustainability, Smart Cities and Infrastructure.

While some Technological Trends, includes: Digital Transformation, Remote Work and Virtual Teams, Gamification and Esports, Extended Reality (XR) and Metaverse, Human-Computer Interaction (HCI). Some of the emerging technologies are: Quantum Computing, Nanotechnology, Biotechnology and Bioinformatics, Synthetic Intelligence and Swarm Intelligence

The Social Impact of diverse and growing computers can be seen in the followings: Digital Divide and Inclusion, AI Ethics and Bias, Cybersecurity Awareness, Online Safety and Harassment and Technology Addiction.

The New Innovations as a growth in the Computing includes:

1. 3D Printing and Additive Manufacturing
2. Autonomous Drones
3. Self-Driving Cars
4. Brain-Computer Interfaces (BCIs)
5. Smart Homes and Buildings

These aforementioned new innovations showcase the diversity and growth in the computer industry.

Digital Applications: Digital applications refer to software programs or platforms that run on digital devices such as computers, smartphones, tablets, or other electronic devices. They can be web-based, mobile-based, or desktop-based, and are designed to perform specific tasks or provide various services.

This encompasses various types of software, platforms, and tools that serve diverse purpose. Digital Applications can be seen in the aspect of Productivity in some as it is used in Microsoft Office, Google Workspace (Docs, Sheets, Slides), Apple Productivity Apps (Pages, Numbers, Keynote), Trello, Asana, Evernote, Dropbox and Slack

While digital applications importance, also be in **Communication** in the following ways such as Email clients (Gmail, Outlook), Messaging apps (WhatsApp, Facebook Messenger), Video conferencing tools (Zoom, Skype, Google meets), Social Media platforms (Facebook, Twitter, LinkedIn, Instagram, Telegram, WhatsApp, Facebook and Messengers), Collaboration tools (Microsoft Teams, Slack).

For **Entertainment** purpose, digital applications can function in areas, such as

1. Streaming services (Netflix, YouTube, Hulu)
2. Music platforms (Spotify, Apple Music)
3. Gaming consoles (PlayStation, Xbox)
4. Mobile games (Pokémon Go, Candy Crush)
5. Virtual reality (VR) and augmented reality (AR) experiences

In Education:

1. Learning management systems (LMS) like Canvas, Blackboard
2. Online course platforms (Coursera, Udemy)
3. Educational apps (Duolingo, Khan Academy)
4. Digital textbooks and e-books
5. Virtual classrooms and webinars

In Health and Wellness:

1. Fitness trackers (Fitbit, Apple Watch)

2. Health apps (MyFitnessPal, Headspace)
3. Telemedicine platforms (Teladoc, American Well)
4. Medical records management (Epic Systems)
5. Mental health support apps (Calm, BetterHelp)

In the Banking and Finance:

1. Mobile banking apps (Bank of America, Chase)
2. Digital payment platforms (PayPal, Venmo)
3. Investment apps (Robinhood, Fidelity)
4. Accounting software (QuickBooks, Xero)
5. Cryptocurrency exchanges (Coinbase, Binance)

In the field of Art and Creative, Digital Applications features in the followings:

1. Graphic design software (Adobe Creative Cloud)
2. Video editing tools (Adobe Premiere, Final Cut Pro)
3. Photo editing apps (Lightroom, Photoshop)
4. Music production software (Ableton, Logic Pro)
5. Writing and publishing platforms (Medium, WordPress)

For Safety and Security of Computer Users, Digital Applications can also be useful as:

1. Antivirus software (Norton, McAfee)
2. Password managers (LastPass, 1Password)
3. VPNs (ExpressVPN, NordVPN)
4. Firewall software
5. Identity theft protection (LifeLock)

Computer System Utilities are vital in the smooth processing and execution of programs/operations, thus Digital applications usefulness in the File management tools (Google Drive, Dropbox), System cleaning and optimization software (CCleaner), Backup and recovery tools (Acronis, Backblaze), Network monitoring software and Weather and news Apps

INFORMATION PROCESSING AND ITS ROLES IN THE SOCIETY:

What is Information Processing?

Information processing refers to the collection, storage, retrieval, manipulation, and dissemination of information using various technologies, such as computers, software, and communication networks.

Information processing plays a vital role in modern society and its impact is multifaceted. It has transformed modern society and its impact will continue to evolve. Addressing the

challenges and concerns associated with information processing is crucial to ensuring its benefits are equitably distributed and its negative consequences mitigated.

Roles of Information Processing.

1. **Decision-making:** Accurate information processing enables informed decisions in various sectors, including business, healthcare, and governance.
2. **Communication:** Effective information processing facilitates communication, collaboration, and knowledge sharing.
3. **Education:** Information processing supports learning, skill development, and access to knowledge.
4. **Innovation:** Processing information drives research, development, and innovation.
5. **Economic growth:** Efficient information processing contributes to economic growth, competitiveness, and job creation.
6. **Healthcare:** Accurate information processing improves healthcare outcomes, disease diagnosis, and treatment.
7. **Governance:** Information processing enables transparent governance, public engagement, and policy-making.
8. **Research and Development:** Facilitates scientific research, data analysis, and innovation.

Some of the Important Positive, Negative and Social Impacts of Information processing, includes the followings:

Positive Impacts:

1. Increased efficiency and productivity
2. Improved accessibility and connectivity
3. Enhanced collaboration and knowledge sharing
4. Economic growth and job creation
5. Better decision-making and problem-solving

Negative Impacts:

1. Information overload and noise
2. Data privacy and security concerns
3. Social isolation and decreased human interaction
4. Dependence on technology
5. Job displacement and automation

The Social impacts are:

1. Accessibility: Information processing makes information accessible to a wider audience.
2. Connectivity: Global connectivity fosters international collaboration and cultural exchange.
3. Empowerment: Access to information empowers individuals, promoting autonomy and self-determination.
4. Social change: Information processing facilitates social movements, advocacy, and activism.
5. Employment: The information processing industry creates jobs and drives economic growth.
6. Privacy and security: Information processing raises concerns about data privacy and security.

The followings are the Information Processing Professionals:

1. Data Analysts
2. Software Developers
3. Information Architects
4. Data Scientists
5. IT Managers
6. Cybersecurity Experts
7. Network Administrators
8. Database Managers
9. Artificial Intelligence/Machine Learning Engineers
10. Information Systems Managers

Some key Industries where the use of Information processing can not be over emphasized, includes:

1. Technology And Software
2. Finance And Banking
3. Healthcare And Biotechnology
4. Education And Research
5. Government And Public Sector
6. Media And Entertainment
7. E-Commerce And Retail
8. Manufacturing And Logistics
9. Energy And Utilities
10. Non-Profit And Social Impact Organizations

The Emerging Trends in the area of Information processing includes:

1. Artificial Intelligence (AI)
2. Cloud Computing
3. Internet of Things (IoT)
4. Blockchain and Distributed Ledger Technology
5. Quantum Computing
6. Extended Reality (XR)
7. 5G Networks
8. Cybersecurity and Threat Intelligence
9. Data Analytics and Visualization
10. Autonomous Systems

The information processing sector continues to evolve, driving innovation and transformation across various industries and aspects of society in the underlisted areas:

(A) Hardware and Architecture:

1. Artificial Intelligence (AI) processors
2. Quantum Computing
3. Neuromorphic Computing
4. Graphene-based Computing
5. 3D Stacked Processing
6. Heterogeneous System Architecture (HSA)
7. High-Performance Computing (HPC)

(B) Software and Programming:

1. Machine Learning (ML) frameworks
2. Deep Learning algorithms
3. Natural Language Processing (NLP)
4. Internet of Things (IoT) development
5. Cloud Computing platforms
6. Containerization (e.g., Docker)
7. Serverless Computing.

There are several **Types of Information Processing** based on the following criteria:

(A) **Cognitive Information Processing:**

1. Perception: Interpreting sensory information.
2. Attention: Focusing on relevant information.
3. Memory: Encoding, storing, and retrieving information.
4. Learning: Acquiring new knowledge and skills.

5. Language Processing: Understanding and generating language.
6. Problem-Solving: Identifying and resolving problems.
7. Decision-Making: Selecting from available options.

(B) Computational Information Processing:

1. Data Processing: Manipulating and transforming data.
2. Algorithmic Processing: Executing step-by-step instructions.
3. Parallel Processing: Processing multiple tasks simultaneously.
4. Distributed Processing: Processing tasks across multiple systems.
5. Cloud Computing: Processing data remotely through cloud services.

(C) Neural Information Processing:

1. Artificial Neural Networks (ANNs): Modeling human brain function.
2. Deep Learning: Multilayer neural networks for complex tasks.
3. Natural Language Processing (NLP): Processing human language.
4. Image Processing: Interpreting and manipulating visual data.
5. Signal Processing: Analyzing and manipulating signals.

(D) Sensory Information Processing: -

1. Visual Processing: Interpreting visual information.
2. Auditory Processing: Interpreting sound information.
3. Tactile Processing: Interpreting touch information.
4. Olfactory Processing: Interpreting smell information.
5. Gustatory Processing: Interpreting taste information.

Other Types are

1. Emotional Processing: Recognizing and managing emotions.
2. Social Information Processing: Interpreting social cues.
3. Spatial Processing: Understanding spatial relationships.
4. Temporal Processing: Understanding time and sequence.
5. Multimodal Processing: Integrating multiple information sources.

The important Theories and Models governing the modern information Processing includes:

1. Information Processing Theory (IPT)
2. Cognitive Load Theory (CLT)
3. Working Memory Model (WMM)
4. Attention Restoration Theory (ART)
5. Global Workspace Theory (GWT)

An Information Processing System (IPS) consists of several **components** that work together to process and manage information. These components interact and work together to form a comprehensive Information Processing System. (IPS). These are the key components:

(A) Hardware Components:

1. Input Devices: Keyboard, mouse, scanner, microphone, etc.
2. Central Processing Unit (CPU): Brain of the computer, executes instructions.
3. Memory (RAM): Temporary storage for data and programs.
4. Storage Devices: Hard disk, solid-state drive, flash drive, etc.
5. Output Devices: Monitor, printer, speaker, etc.

(B) Software Components:

1. Operating System (OS): Manages hardware and software resources.
2. Application Software: Programs that perform specific tasks (e.g., word processing).
3. Utility Software: Programs that manage and maintain the system (e.g., antivirus).
4. Firmware: Permanent software stored in non-volatile memory.

(C) Data Components:

1. Input Data: Raw data entered into the system.
2. Processed Data: Transformed data after processing.
3. Output Data: Resultant data after processing.
4. Stored Data: Permanent data stored in storage devices.

(D) Process Components:

1. Input Process: Capturing and converting input data.
2. Processing: Executing instructions and transforming data.
3. Output Process: Presenting processed data.
4. Storage Process: Storing and retrieving data.

(E) Communication Components:

1. Network Interface: Connecting to other systems or networks.
2. Communication Protocols: Rules governing data exchange.
3. Data Transmission: Sending and receiving data.

(F) Human Components:

1. User: Interacting with the system.
2. Operator: Managing and maintaining the system.
3. Analyst: Designing and developing the system.

Other Components:

1. Database Management System (DBMS): Managing and Storing data.
2. Knowledge Base: Storing and retrieving knowledge.
3. Decision Support System (DSS): Supporting decision-making process.
4. Artificial Intelligence (AI) and Machine Learning (ML) components.

The Internet: Its Application and Impact on the World Today

Introduction

The internet's fascinating history began in the 1960s, when the United States Department of Defense launched the Advanced Research Projects Agency Network (ARPANET), connecting four computers to facilitate communication between government and academic researchers. This pioneering project laid the groundwork for the modern internet. In the 1970s, the Internet Protocol (IP) was developed, enabling different networks to communicate and paving the way for the internet's expansion. The 1980s marked a significant milestone in the internet's evolution, with the introduction of the Domain Name System (DNS), which simplified website addresses, and Internet Relay Chat (IRC), which enabled real-time online communication. The World Wide Web (WWW), invented by Tim Berners-Lee in 1989, revolutionized the internet by making content accessible to a broader audience.

During the 1990s, there was rapid growth in internet development, as dial-up internet brought connectivity to homes, email transformed communication, and search engines like Yahoo! and Google simplified information retrieval. E-commerce emerged, allowing online shopping to become a staple of modern life. In the 2000s, social media platforms like Facebook, Twitter, and Instagram redefined social interactions, while mobile internet and smartphones made access ubiquitous. Cloud computing enabled remote data storage and processing, and big data analytics provided valuable insights from vast amounts of data.

Today, the internet's impact on the world is profound. On the positive side, it has bridged geographical divides, democratized knowledge, driven economic growth, and transformed education and healthcare. However, it also poses significant challenges, including cybersecurity threats, social isolation, misinformation, and environmental concerns. The next subsection will explore the application and impact of the internet in detail.

The Internet: Its Application and Impact

Revolution of Communication

The internet has transformed various industries, including finance, media, healthcare, education, and governance. The internet has revolutionised modern life, seamlessly integrating into various aspects of daily routines. Its impact is profound, transforming the way we

communicate, work, learn, and entertain ourselves. To begin with, communication has been redefined by the internet. Social media platforms like Facebook, Twitter, Instagram, and LinkedIn have connected people worldwide, fostering global relationships and communities. Email has become an essential tool for personal and professional communication, while video conferencing services like Zoom, Skype, and Google Meet enable virtual meetings and remote collaborations. Messaging apps like WhatsApp, WeChat, and Telegram facilitate instant communication, bridging geographical divides.

Educational Advancements

In another development, the internet has transformed education, making learning more accessible and convenient. Online courses and degree programs are available through platforms like Coursera, Udemy, and edX. Virtual classrooms enable remote learning, while digital resources like e-books, academic journals, and educational websites provide invaluable information. Online tutoring services offer personalized learning experiences, catering to individual needs.

Arts and Entertainment

Entertainment has also been transformed by the internet. Streaming services like Netflix, YouTube, Amazon Prime, and Hulu provide endless options for movies, TV shows, and music. Online gaming has become a vibrant industry, with multiplayer games, esports, and virtual reality experiences captivating audiences. Music and podcast streaming services like Spotify, Apple Music, and podcasts offer diverse content, while online communities and forums connect people with shared interests.

Health and Social Care

The domain of health and social care has equally benefited from the internet. Telemedicine enables virtual consultations and remote healthcare services, expanding access to medical care. Online health information resources provide valuable insights for medical research and health education. Electronic health records and secure data storage ensure confidential and efficient healthcare management. Wearable devices and mobile apps track vital signs, promoting health monitoring and wellness.

Fintech

The finance sector has been transformed by the internet. Online banking enables mobile banking, digital payments, and financial transactions. Cryptocurrency, including Bitcoin and Ethereum, has emerged as a digital alternative to traditional currency. Mobile wallets like Apple Pay, Google Pay, and PayPal facilitate secure transactions. Investment platforms offer online trading, brokerage services, and financial analysis.

Transportation and Logistics

Transportation and logistics companies are being transformed by the Internet. Ride-hailing services like Uber and Lyft have revolutionized urban mobility. Food delivery platforms like GrubHub and UberEats have made ordering and delivery seamless. Navigation services like Google Maps and Waze provide real-time traffic updates and directions. Supply chain management has been optimized with digital tracking, logistics, and inventory management. Even in developing countries such as Nigeria, various countries are taking advantage of internet enablers to achieve real time logistics services.

The World of Work

In many ways, government and public services have also been impacted by the internet. E-government portals provide citizens with easy access to services, licenses, and permits. Digital democracy initiatives enable online voting systems, public forums, and participatory governance. Public records are now accessible online, promoting transparency and accountability. Emergency services have been enhanced with online reporting, emergency response systems, and disaster management.

Broadly speaking, the internet has significantly impacted the world of work and business. Remote work arrangements have become increasingly popular, with telecommuting and virtual offices enabling flexible work environments. E-commerce has thrived, with online shopping, digital payments, and entrepreneurship on the rise. Digital marketing has emerged as a crucial aspect of business strategy, with online advertising, social media marketing, and SEO driving business growth. Cloud computing has enabled remote data storage, processing, and collaboration.

Conclusion

The internet's evolution has been a remarkable journey, transforming modern society and redefining the way we communicate, access information, and interact with the world. As the internet evolves, its potential to shape the future of humanity remains limitless. The internet's potential for positive impact remains immense, and its advancement continues to shape the world. Its applications continue to expand, driving innovation, progress, and global connectivity.

The Different Areas/Programs of the Computing Discipline

Introduction

Instead of being a single subject, computing is a collection of disciplines with diverse areas of interest. Computing is any goal-oriented activity that makes use of, benefits from, or creates computers. Therefore, computing comprises the design and development of computer hardware and software components for a range of applications, such as processing, organizing, and managing different types of information; conducting computer-intensive scientific research; creating and utilizing communications and entertainment media; finding and compiling information relevant to a particular objective, among other things.

There is an endless variety of applications for computers and the field of computing has undergone tremendous growth and diversification in recent years, transforming the way we live, work, and interact. Computing discipline encompasses a broad range of specialties, each with its unique focus, methodologies, and applications. Understanding these areas is crucial for navigating the complex landscape of modern technology. This discussion aims to provide an overview of the various areas of computing discipline, explaining each in easy-to-understand language. By exploring these fields, we can gain insight into the exciting developments and innovations shaping our digital world.

Areas/Programs of the Computing Discipline

The computing discipline is made up of different domains that are interconnected and constantly evolving. The various areas/programmes of computing are responsible towards driving innovation and technological advancements. They are identified and discussed below:

Computer Networks

Computer Networks involve connecting devices to share resources and communicate. This includes:

- Local Area Networks (LANs): Connecting devices within a building.
- Wide Area Networks (WANs): Connecting devices across cities or countries.
- Internet of Things (IoT): Connecting everyday devices to the internet.

Software Engineering

Software Engineering develops reliable, efficient software. This involves:

- Design Patterns: Reusing proven solutions.
- Testing: Ensuring software quality.
- Version Control: Managing software changes.

Web Development

Web Development builds websites and web applications. This includes:

- Front-end Development: Creating user interfaces.
- Back-end Development: Building server-side logic.
- Web Design: Crafting visually appealing websites.

Artificial Intelligence (AI)

Artificial Intelligence focuses on creating intelligent machines that can think, learn, and act like humans. AI involves:

- Machine Learning: Teaching computers to learn from data.
- Natural Language Processing: Enabling computers to understand human language.
- Robotics: Building robots that can interact with their environment.

Cybersecurity

Cybersecurity protects computer systems and data from unauthorized access. This includes:

- Encryption: Securing data with codes.
- Firewalls: Blocking unauthorized access.
- Threat Detection: Identifying potential security risks.

Data Science

Data Science extracts insights from large datasets. This involves:

- Data Mining: Discovering patterns in data.
- Data Visualization: Presenting data in a clear format.
- Statistical Analysis: Interpreting data using statistics.

Database Systems

Database Systems manage and store data efficiently. This includes:

- Database Design: Creating structured databases.
- Data Retrieval: Accessing data quickly.
- Data Security: Protecting sensitive data.

Human-Computer Interaction (HCI)

Human-Computer Interaction designs user-friendly interfaces. This involves:

- User Experience (UX) Design: Creating intuitive interfaces.
- User Interface (UI) Design: Building visually appealing interfaces.
- Accessibility: Ensuring technology is usable by everyone.

Information Systems

Information Systems integrate technology into organizations. This includes:

- Business Intelligence: Analysing data for business decisions.
- Enterprise Resource Planning (ERP): Managing business operations.
- Supply Chain Management: Coordinating logistics.

Operating Systems

Operating Systems manage computer hardware resources. This includes:

- Process Management: Managing running programs.

- Memory Management: Allocating system memory.
- File Systems: Organizing stored data.

Conclusion

From the foregoing, it is clear that the computing discipline encompasses a rich tapestry of specialized fields, each driving innovation and progress in its unique way. From Artificial Intelligence to Web Development, these areas have transformed industries, revolutionized communication, and improved lives. As technology continues to evolve, understanding these computing disciplines is essential for harnessing their potential and addressing the challenges of the digital age. By recognizing the interconnectedness of these fields, we can foster collaboration, inspire new breakthroughs, and shape a brighter future for all. As we move forward, the possibilities offered by computing discipline will continue to expand, empowering us to create, innovate, and thrive in an increasingly digital world.

THE JOB SPECIALIZATIONS FOR COMPUTING SCIENCE PROFESSIONALS

1 **Software engineer:** is a professional who designs develops, tests and maintains software application

2 **Cyber security analyst:** is a professional responsible for protecting an organization's computer systems and networks from cyber threats, working closely with upper management. The cyber security recommends safeguards to keep information confidential and authorize internal employees to use parts of the network. They also develop procedures to respond to emergencies and restore or backup digital item

3 **A user experience researcher:** is a data expert who analyzes member of target audiences to understand what they look for when using a digital program.

4 **A video game designer:** is a professional responsible for creating visual elements for video games for mobile devices, computers and gaming systems. Using programming languages and graphics design, the video game designer builds characters and settings that coincide with the games storylines and they test the game for functionality, easy navigation and visual appeal. The professional works closely with animators and programmers to build the game and they strategize ways to advertise it to encourage consumers to purchase it.

5 **A business intelligence analyst:** Is a professional who evaluates the operations of a company to identify ways to make it more successful. With expertise in data science, the business intelligence analyst determines if the company is making progress toward its goals by assessing the resources. It uses and the challenges it's faced. Another responsibility is performing competitor analysis, which helps the professional stay informed about the industry and develop strategies to exceed competing business.

6 **A data base Administrator:** is a professional who oversees activities in software database that a company uses to store and organize information such as user login credentials, client interactions and survey results to maintain the confidentiality of the records the database administer ensures the structures are working effectively and they install security procedures to identify threats, remove viruses and restore lost data. The administrator may also install updates on the databases to boost their performance and expand their capability

7 **A frontend developer:** is a programmer who specializes in the design and implementation of the user interface (UI) of websites and applications: They primarily focus on the part of a website that users interact with directly using language such as HTML, CSS and JavaScript frontend developers ensure that the visual elements are responsive and functional, creating a seamless and engaging user experience. They also collaborate with designers and back-end developers to integrate and optimize web applications.

8 Full stack developer: is a software developer who is proficient in both front-end and back-end development. They work on all layers of a web applications including the user interface (UI) server database and server-side applications. Full stack developers are skilled in various technologies and programming language enabling them to handle all aspect of development from designing the user experience to managing databases and server logic their versatility allow them to contribute to both the client-side and server-side of application

9 A mobile application developer: is a software engineer who specializes in creating applications for mobile devices such as smartphones and tablets They design, build, test and maintain mobile apps for various platforms primarily iOS and Android mobile developers use programming languages like swift, kotlin, or Java and frameworks like React Native or flutter to develop user-friendly, functional and efficient mobile applications.

10 System engineer: is an industry expert who creates a process for conceptualizing, developing and implementing a system, such as a new software application or piece of computer hardware. To maximize efficiency for the process the systems engineer compiles a list of necessary resources collaborating with professionals and establishing parameters to evaluate the success of the project. They also prioritize the safety and security of their products and lend technical expertise to assist other technology specialists on their team.

11 A network security engineer: is an IT professional who installs safeguards to protect a computer network from harm which can include viruses and malware. Network security engineers analyze the performance of the computer to identify malfunctions and prevent them recurring, and they conduct tests to see how vulnerable the network is to external threats.

THE FUTURE OF COMPUTING

Computer technology has come a long way in the past few decades. The world of computing is ever-changing and growing at an exponential rate; it would be impossible to predict all the advances in the field that are sure to occur in upcoming years. However, there are certain key areas where experts anticipate significant progress. One example is Artificial intelligence(AI). AI has already begun to make its mark in various sectors from healthcare to finance-but this is just the beginning. Experts believe that AI will soon become ubiquitous across industries as well as everyday lives.

Another trend expected to continue into the near future is cloud computing. It is a form of distributed computing where resources are shared among many users who access them through the internet. This type of technology has enabled businesses and individuals to store data online and make use of powerful analysis tools without having to purchase hardware and software.

Cloud computing has enabled businesses and individuals alike to access data faster than ever before while reducing their need for physical storage space. Going forward, cloud computing is

likely to become even more popular and prevalent due to technological advancements like 5G networks providing increased speed and reliability

Over the coming ten years, improvement in personal computer technology are anticipated to concentrate on better user experience boosting connectivity and improving performance. The widespread use of artificial intelligence and machine learning, the creation of even faster and more effective processors and the incorporation of virtual and augmented reality technologies into personal computing are all prediction for the future of personal computer.

Machine learning and deep learning are two of the most powerful emerging technologies in computer technology. These methods allow computers to learn from data, identify patterns and make prediction without being explicitly programmed. This has applications across many industries including healthcare, finance, retail, education and more.

The potential for machines learning and deep learning is immense. With their ability to analyze vast amounts of data quickly and accurately. They could revolutionize the way we interact with machines. For example, they can be used to build smarter robots that understand instructions better or medical AI system that accurately diagnose diseases. As these technologies become more advanced over time, their impact on our lives will only increase.

In addition to this potential growth in capabilities machine learning and deep learning also offer a cost-efficient solution for business looking to stay ahead of the competition. By freeing up resources previously spend on manual tasks by automating them via ML/DL algorithms, companies can save money while still achieving highly accurate results. It's clear why so many tech giants are investing heavily in the development of these two promising technologies. The possibilities seem endless.

Additionally, it is anticipated that new technologies like 5G and the internet of things(IoT) will have a big impact on personal computers.

The internet of things (IoT) is a quickly growing technology that links physical objects to the internet. It involves connecting everyday items such as light bulbs, washing machines and security systems together so they can interact with each other in real-time or on an automated cycle.

This means we will be able to control our home appliances remotely from anywhere in the world without having to be physically present. This could save us time, energy and money by allowing us to manage devices efficiently even when we are not at home IoT also has potential applications for healthcare, smart cities and connected vehicles.

Personal computers are going to become more ingrained in our daily lives, acting as hubs for managing and controlling various devices and services as smart devices and connected homes become more common place.

Quantum computers have the potential to greatly improve computing even at this early stage of development. In addition, they will be able to solve complex problems that traditional computers currently cannot solve.